1 Gain law

(bars denote normalized quantities: lengths by R, $\bar{a} = a/a_o$, $\bar{f} = a_o f$; $0 < \bar{a} < 1$; $e_\theta = \theta - \hat{\theta}$; b is a FIXED constant here, not a state)

$$\bar{a}_{\text{true}}(\bar{w}) = 1 - \frac{9}{8\bar{w}} + \frac{1}{2\bar{w}^3} - \frac{57}{100\,\bar{w}^4} + \frac{1}{5\bar{w}^5} + \frac{7}{200\,\bar{w}^{11}} - \frac{1}{25\,\bar{w}^{12}}$$

$$\bar{a}'_{\text{true}}(\bar{w}) > 0 \;\text{ on }\; \bar{w} > 1 \qquad \Longrightarrow \qquad \bar{a}' = b_{\text{true}}(\bar{a})\big(1-\bar{a}\big)^2 \;\text{ exactly,} \qquad b_{\text{true}} := \frac{\bar{a}'_{\text{true}}}{\big(1-\bar{a}_{\text{true}}\big)^2}$$

$$\begin{aligned} \bar{a}_{\text{true}}(1) = 0, \quad \bar{a}'_{\text{true}}(1) = 1 \quad &\Longrightarrow \quad b_{\text{true}}\big(\bar{a} = 0\big) = 1 \\ 1 - \bar{a}_{\text{true}} = \frac{9}{8\bar{w}} + \mathcal{O}(\bar{w}^{-3}) \quad &\Longrightarrow \quad b_{\text{true}}\big(\bar{a} \to 1\big) = \frac{8}{9} \end{aligned}$$

$$\begin{aligned} &\bar{a}'(\bar{a}) := b\big(1-\bar{a}\big)^2, \qquad b = \tfrac{8}{9} \text{ fixed} \\ &\quad a_w = a_o\,\bar{a}_w, \qquad a_o = \frac{\Delta t}{\gamma_N R} \end{aligned}$$

2 Jacobians

$$\begin{aligned} \frac{\partial \bar{a}'}{\partial \bar{a}} &= -2b\big(1-\bar{a}\big) \\ \frac{\partial \bar{a}'}{\partial \bar{w}_D} &= 0 \\ \frac{\partial \bar{a}_w[k+1]}{\partial \bar{w}_D[k]} &= \bar{a}' = b\big(1-\bar{a}_w\big)^2 \end{aligned}$$

3 True dynamics

$\big(\delta\bar{w}_j[k] := \delta\bar{w}[k-d+j-1],\; j = 1,\ldots,d+1,\; d = 2\big)$

$$\bar{w}_D[k] = \bar{w}_D^0[k] + (1-\lambda_c)\sum_{i=1}^{d} \bar{w}_D^0[k-i]$$

$$
\begin{aligned}
\bar{w}[k] &= \bar{w}_d[k] - \delta\bar{w}_3[k] \\
\bar{w}_d[k+1] &= \bar{w}_d[k] + \Delta\bar{w}_d[k] \\
\delta\bar{w}_1[k+1] &= \delta\bar{w}_2[k] \\
\delta\bar{w}_2[k+1] &= \delta\bar{w}_3[k] \\
\delta\bar{w}_3[k+1] &= \lambda_c\,\delta\bar{w}_3[k] - F_{dw}[k]\,e_{\bar{a}_w}[k] - e_{\bar{w}_D}[k] - \varepsilon_w[k] \\
\bar{a}_w[k+1] &= \bar{a}_w[k] + \bar{a}'_{\text{true}}[k]\Big[\Delta\bar{w}_d[k] + (1-\lambda_c)\,\delta\bar{w}_3[k] + F_{dw}[k]\,e_{\bar{a}_w}[k] + e_{\bar{w}_D}[k] + \varepsilon_w[k]\Big] \\
\bar{w}_D[k+1] &= \bar{w}_D[k] + \delta\bar{w}_D[k] \\
\delta\bar{w}_D[k+1] &= \delta\bar{w}_D[k]
\end{aligned}
$$

$$
\begin{aligned}
F_{dw}[k] &= \bar{f}_{dw}[k] + (1-\lambda_c)\sum_{i=1}^{d}\bar{f}_{dw}[k-i] \\
\varepsilon_w[k] &= \bar{a}_w[k]\,\bar{f}_T[k] + (1-\lambda_c)\sum_{i=1}^{d}\bar{a}_w[k-i]\,\bar{f}_T[k-i] + (1-\lambda_c)\,n_w[k]
\end{aligned}
$$

4 Process model

$$
\mathbf{x} = \big[\,\delta\bar{w}_1,\ \delta\bar{w}_2,\ \delta\bar{w}_3,\ \bar{a}_w,\ \bar{w}_D,\ \delta\bar{w}_D\,\big]^\top \qquad (d=2)
$$

5 Estimator

$$
\begin{aligned}
\delta\hat{\bar{w}}_1[k+1] &= \delta\hat{\bar{w}}_2[k] \\
\delta\hat{\bar{w}}_2[k+1] &= \delta\hat{\bar{w}}_3[k] \\
\delta\hat{\bar{w}}_3[k+1] &= \lambda_c\,\delta\hat{\bar{w}}_3[k] \\
\hat{\bar{a}}_w[k+1] &= \hat{\bar{a}}_w[k] + b\big(1-\hat{\bar{a}}_w[k]\big)^2\Big[\Delta\bar{w}_d[k] + (1-\lambda_c)\,\delta\hat{\bar{w}}_3[k]\Big] \\
\hat{\bar{w}}_D[k+1] &= \hat{\bar{w}}_D[k] + \delta\hat{\bar{w}}_D[k] \\
\delta\hat{\bar{w}}_D[k+1] &= \delta\hat{\bar{w}}_D[k]
\end{aligned}
$$

6 Error dynamics

(True − Estimator; coefficients at the estimate)

$$
\begin{aligned}
e_{\delta\bar{w}_1}[k+1] &= e_{\delta\bar{w}_2}[k] \\
e_{\delta\bar{w}_2}[k+1] &= e_{\delta\bar{w}_3}[k] \\
e_{\delta\bar{w}_3}[k+1] &= \lambda_c\, e_{\delta\bar{w}_3}[k] - F_{dw}[k]\, e_{\bar{a}_w}[k] - e_{\bar{w}_D}[k] - \varepsilon_w[k] \\
e_{\bar{a}_w}[k+1] &= \Big[1 + b\big(1-\hat{\bar{a}}_w\big)^2 F_{dw} - 2b\big(1-\hat{\bar{a}}_w\big)\big(\Delta\bar{w}_d + (1-\lambda_c)\delta\hat{\bar{w}}_3\big)\Big] e_{\bar{a}_w}[k] \\
&\quad + (1-\lambda_c)\, b\big(1-\hat{\bar{a}}_w\big)^2 e_{\delta\bar{w}_3}[k] + b\big(1-\hat{\bar{a}}_w\big)^2 e_{\bar{w}_D}[k] \\
&\quad + b\big(1-\hat{\bar{a}}_w\big)^2 \varepsilon_w[k] \\
e_{\bar{w}_D}[k+1] &= e_{\bar{w}_D}[k] + e_{\delta\bar{w}_D}[k] \\
e_{\delta\bar{w}_D}[k+1] &= e_{\delta\bar{w}_D}[k]
\end{aligned}
$$

$$
F_e[k] = \begin{bmatrix}
0 & 1 & 0 & 0 & 0 & 0 \\
0 & 0 & 1 & 0 & 0 & 0 \\
0 & 0 & \lambda_c & -F_{dw} & -1 & 0 \\
0 & 0 & (1-\lambda_c)\, b\big(1-\hat{\bar{a}}_w\big)^2 & \begin{matrix} 1 + b\big(1-\hat{\bar{a}}_w\big)^2 F_{dw} \\ -2b\big(1-\hat{\bar{a}}_w\big)\big(\Delta\bar{w}_d + (1-\lambda_c)\,\delta\hat{\bar{w}}_3\big) \end{matrix} & b\big(1-\hat{\bar{a}}_w\big)^2 & 0 \\
0 & 0 & 0 & 0 & 1 & 1 \\
0 & 0 & 0 & 0 & 0 & 1
\end{bmatrix}
$$

$$
\mathbf{q}[k] = \Big[0,\ 0,\ -\varepsilon_w,\ b\big(1-\hat{\bar{a}}_w\big)^2 \varepsilon_w,\ 0,\ 0 \Big]^\top
$$

7 Measurement and innovation

$$
\begin{aligned}
\nabla_d \bar{w}_d[k] &= \bar{w}_d[k] - \bar{w}_d[k-d] \\
y_1[k] &= \delta\bar{w}_1[k] + n_w[k] \\
y_2[k] &= \bar{a}_w[k-d] + n_a[k]
\end{aligned}
$$

$$
\bar{a}_w[k-d] = \bar{a}_w[k] - b\big(1-\bar{a}_w[k]\big)^2 \nabla_d \bar{w}_d[k]
$$

$$
\begin{aligned}
\frac{\partial y_2}{\partial \bar{a}_w} &= 1 + 2b\big(1-\hat{\bar{a}}_w\big)\nabla_d \bar{w}_d \\
\frac{\partial y_2}{\partial \bar{w}_D} &= 0 \\
\frac{\partial y_2}{\partial \delta\bar{w}_D} &= 0
\end{aligned}
$$

$$H[k] = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 + 2b(1-\hat{\bar{a}}_w)\nabla_d \bar{w}_d & 0 & 0 \end{bmatrix}$$

$$L[k] = P^f H^\top \left(H P^f H^\top + R \right)^{-1}$$

$$\begin{aligned} e_{y_1}[k] &= e_{\delta\bar{w}_1}[k] + n_w[k] \\ e_{y_2}[k] &= \left[1 + 2b\left(1-\hat{\bar{a}}_w\right)\nabla_d \bar{w}_d \right] e_{\bar{a}_w}[k] + n_a[k] \end{aligned}$$

8 Process and measurement noise

$$\begin{aligned} q_3 &= -\bar{a}_w \, \bar{f}_T \\ q_4 &= -b\left(1-\hat{\bar{a}}_w\right)^2 q_3 \\ \gamma(\bar{w}) &= \gamma_N c(\bar{w}) \\ \sigma^2_{\bar{f}_T} &= a_o^2 \, \frac{4k_B T \gamma(\bar{w})}{\Delta t} \end{aligned}$$

$$\begin{aligned} Q_{33} &= \frac{4k_B T \, a_o}{R} \, \bar{a}_w \\ Q_{34} = Q_{43} &= -b\left(1-\hat{\bar{a}}_w\right)^2 Q_{33} \\ Q_{44} &= b^2\left(1-\hat{\bar{a}}_w\right)^4 Q_{33} \\ Q_{55} = Q_{66} &= 0 \end{aligned}$$

$$\begin{aligned} R &= \mathrm{diag}(R_1, R_2) \\ R_1 &= \sigma^2_{n_w} \\ R_2 &= K_{\mathrm{var}} I F_{\mathrm{var}} \left(\bar{a}_w + \xi\right)^2 + d\, Q_{44} \\ K_{\mathrm{var}} &= \frac{2a_{\mathrm{cov}}}{2 - a_{\mathrm{cov}}} \\ \sigma^2_{\delta\bar{w}_r} &= C_{dpmr} \frac{4k_B T \, a_o}{R} \left(\bar{a}_w + \xi\right) \\ \xi &= \frac{C_n \sigma^2_{n_w}}{C_{dpmr}\left(4k_B T a_o / R\right)} \end{aligned}$$

Seed and prior

$$
\begin{aligned}
\bar{a}(\bar{w}) &= 1 - \frac{1}{b\,(\bar{w} - \bar{w}_s)} \\
\hat{\bar{a}}_w[0] &= 1 - \frac{1}{b\,(\bar{w}[0] - \bar{w}_s)} \\
\hat{\bar{w}}_D[0] &= \delta\,\hat{\bar{w}}_D[0] = 0
\end{aligned}
$$

The container for the pair is the height offset that reproduces a wrong wall law exactly. With the plant on b_p and the estimator on b,

$$
\begin{aligned}
1 - \frac{1}{b_p\,\bar{w}} = 1 - \frac{1}{b\,(\bar{w} + \delta)} \qquad &\Longrightarrow \qquad \delta(\bar{w}) = \bar{w}\left(\frac{b_p}{b} - 1\right) \\
\dot{\delta} = \dot{\bar{w}}\left(\frac{b_p}{b} - 1\right)
\end{aligned}
$$

$$
\begin{aligned}
\kappa_b &:= \sup\left|\frac{b_p}{b} - 1\right| \qquad \text{(physical input: how wrong the wall law may be)} \\
P_{55}[0] &= \Big[\kappa_b \sup_{\bar{w}\in\mathcal{E}} \bar{w}\Big]^2 \\
P_{66}[0] &= \Big[\kappa_b \sup_k \big|\Delta\bar{w}_d[k]\big|\Big]^2 \\
P_{56}[0] &= 0
\end{aligned}
$$

Observability

Gate 1. Both routes for $\bar{w}_D$:

$$
\begin{aligned}
\text{state route} \quad \frac{\partial \delta\bar{w}_3[k+1]}{\partial \bar{w}_D} &= -1, \qquad \frac{\partial \bar{a}_w[k+1]}{\partial \bar{w}_D} = b\big(1 - \hat{\bar{a}}_w\big)^2 \\
\text{measurement route} \quad \frac{\partial y_1}{\partial \bar{w}_D} &= \frac{\partial y_2}{\partial \bar{w}_D} = 0
\end{aligned}
$$

Gate 2. Time signatures on the position channel:

$$
\begin{aligned}
e_{\bar{a}_w} &\longrightarrow -F_{dw}[k]\,e_{\bar{a}_w} && \text{scaled by the applied force} \\
e_{\bar{w}_D} &\longrightarrow -e_{\bar{w}_D} && \text{unit coefficient} \\
e_{\delta\bar{w}_D} &\longrightarrow \text{ramp}
\end{aligned}
$$

Gate 3. Two reductions. Through y_2, with $\bar{w}_D$ constant and $h := 1 + 2b(1 - \hat{\bar{a}}_w)\nabla_d\bar{w}_d$:

$$A_2 = \begin{bmatrix} 1 + b\big(1-\hat{\bar{a}}_w\big)^2 F_{dw} - 2b\big(1-\hat{\bar{a}}_w\big)\big(\Delta\bar{w}_d + (1-\lambda_c)\delta\hat{\bar{w}}_3\big) & b\big(1-\hat{\bar{a}}_w\big)^2 \\ 0 & 1 \end{bmatrix}, \qquad C_2 = \begin{bmatrix} h & 0 \end{bmatrix}$$

$$\det O_2 = \det \begin{bmatrix} C_2 \\ C_2 A_2 \end{bmatrix} = h^2\, b\big(1-\hat{\bar{a}}_w\big)^2 = h^2\, \bar{a}'$$

$$\det O_2 = 0 \iff \bar{a}' = 0 \;\; (\bar{a} \to 1,\ \text{far field}) \quad \text{or} \quad \nabla_d \bar{w}_d = -\frac{1}{2b(1-\hat{\bar{a}}_w)}$$

Through y_1, the channel sees only the combination $F_{dw}[k]\, e_{\bar{a}_w} + e_{\bar{w}_D}$, so two consecutive steps give

$$\det \begin{bmatrix} F_{dw}[k] & 1 \\ F_{dw}[k+1] & 1 \end{bmatrix} = F_{dw}[k] - F_{dw}[k+1]$$

$$\det O_2 = 0 \iff F_{dw}[k+1] = F_{dw}[k] \quad \text{(constant applied force)}$$